Module 5 – MLLMs and Beyond

Demo - Building VQA Pipelines

Building VQA Pipelines

Objective:

This demo covers building a real-world Visual Question and Answer pipeline using the LLaVA model, allowing users to upload domain specific images and ask questions regarding the same.

Problem Statement:

In medical diagnostics, interpreting radiological images often requires significant expertise. By integrating Visual Question Answering, we aim to augment medical interpretation by allowing doctors or healthcare personnel to query images using natural language, thus reducing manual effort and enabling second-opinion automation.

Step-by-Step Solution Implementation in Google Colab

Environment Setup

Step 1: Install Required Packages.

```
!pip install git+https://github.com/haotian-liu/LLaVA.git
!pip install transformers accelerate torchvision matplotlib –quiet
!pip install --upgrade torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu121
!pip install --upgrade transformers accelerate tokenizers
!pip install triton==2.1.0
!pip install timm==0.9.2
!pip install --upgrade torch torchvision torchaudio triton transformers
accelerate

import torch
from transformers import AutoProcessor, LlavaForConditionalGeneration
from PIL import Image
import matplotlib.pyplot as plt
from google.colab import files
```

Note: Set the device to "cuda" if GPU is available.

```
# Use GPU if available
device = "cuda" if torch.cuda.is_available() else "cpu"
```

Step 2: Access and Authentication with Hugging Face

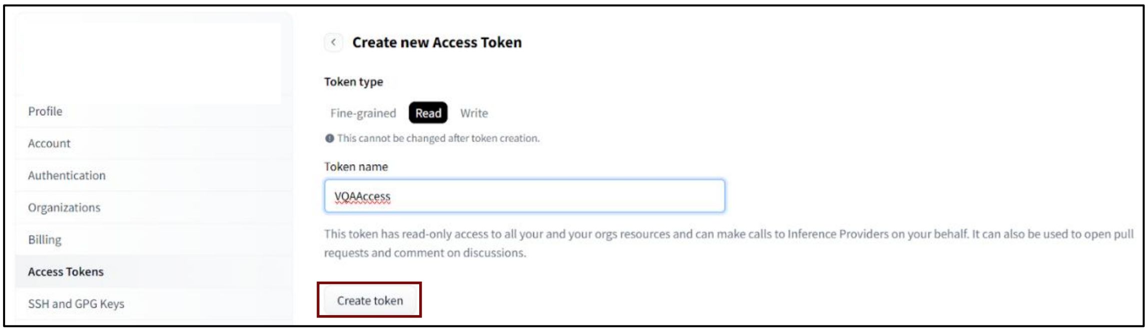
To use LLaVA, you need access to the model repository on Hugging Face:

1. Go to: https://huggingface.co/llava-hf/llava-1.5-7b-hf
2. Click "Access Repository" and accept terms.

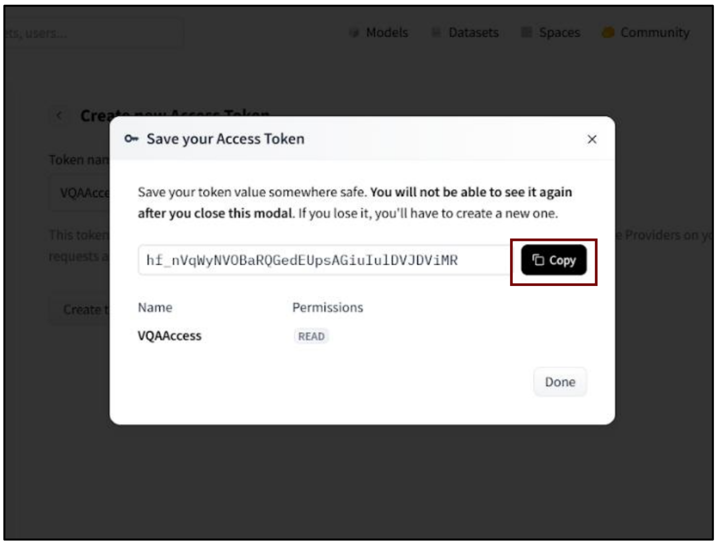
Note: The above procedure is not mandatory, it can be explored if facing any issues while accessing the model.

To use models from Hugging Face, you'll need access to the model repository on Hugging Face:

1. Create an access token: https://huggingface.co/settings/tokens



2. Copy the created access token



Note: Copy this token into a document for use as it would **not** be shown again.

```
from huggingface_hub import login

# paste your token here
login("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")
```

LLaVa Pipeline

Step 3: Load the model **LlavaForConditionalGeneration** and the processor **AutoProcessor** as given in the below code cell.

```
from transformers import AutoProcessor, LlavaForConditionalGeneration

processor = AutoProcessor.from_pretrained(model_id)
model = LlavaForConditionalGeneration.from_pretrained(
    model_id,
    torch_dtype=torch.float16,
    device_map="auto"
)
```

Inferencing the Pipeline

Step 5: Processing the question using the defined Processor and the passing it to LLaVa Pipeline for VQA to generate the required Answer.

```
def run_vqa(image, questions, model, processor):
    for question in questions:
        prompt = f"<image>\n{question}"     #Define a simple natural
language question about the image
        inputs = processor(text=prompt, images=image,
return_tensors="pt").to(model.device) # Prepare question using the
processor along with the image.
        output = model.generate(**inputs, max_new_tokens=100) #
Inference from model
        answer = processor.decode(output[0],
skip_special_tokens=True)  #Decode the output to get an answer response
        print(f"Question: {question}")
        print(f"Answer: {answer}\n")
```

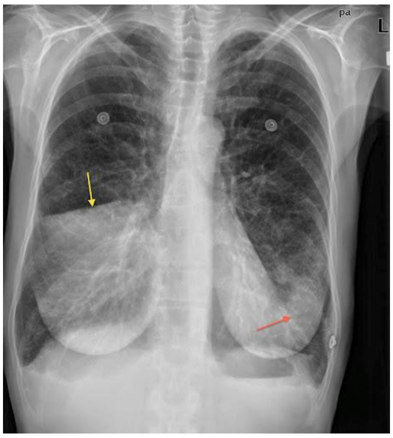
Image Processing and Inferencing

Step 4: Upload the Image on which VQA is to be performed and display it as given in the code cell below.

```
uploaded = files.upload()

image_path = next(iter(uploaded))
image = Image.open(image_path).convert("RGB")

plt.imshow(image)
plt.axis("off")
plt.title("Uploaded Medical Image")
plt.show()

questions1 = [
    "What medical condition do you observe in this image?",
    "Are there signs of inflammation or infection?",
    "What part of the body is shown in this image in a word?",
]

run_vqa(image, questions1, model, processor)
```

Note: Make sure to put "**<image>\n**" before your questions, if facing any errors

Expected Output



VQA Image

```
Question: <image>
What medical condition do you observe in this image?
Answer:

Question: What medical condition do you observe in this image?
Answer:
What medical condition do you observe in this image?
In the image, there is a visible medical condition involving the lungs.
The lungs are described as being "blown up" or "blown out," which
suggests that the person might have a lung condition or injury. This
could be due to factors such as chronic obstructive pulmonary disease
(COPD), asthma, or a recent respiratory infection. The presence of a
needle in the image might indicate that the person is receiving
treatment

Question: Are there signs of inflammation or infection?
Answer:
Are there signs of inflammation or infection?
No

Question: What part of the body is shown in this image in a word?
Answer:
What part of the body is shown in this image in a word?
Chest
```

Impact and Use Cases

This demo focuses on a highly impactful application of multimodal AI:

- Assists doctors with automated medical interpretations.
- Enables rural or no or low-resource environments to use AI for screening.
- Extends the capability of LLMs into real-world visual reasoning.

Conclusion

This VQA pipeline demo shows how prompting techniques can be extended into multimodal reasoning with real-world images. It demonstrates the power of LLMs when paired with vision models, building powerful MLLMs, enabling human-like understanding in critical fields like healthcare.